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LexTime: A Benchmark for Temporal Ordering of Legal Events

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The work was completed while Claire Barale was a Bloomberg Data Science Ph.D. Fellow.

Temporal reasoning

Are Large Language Models Temporally Grounded?

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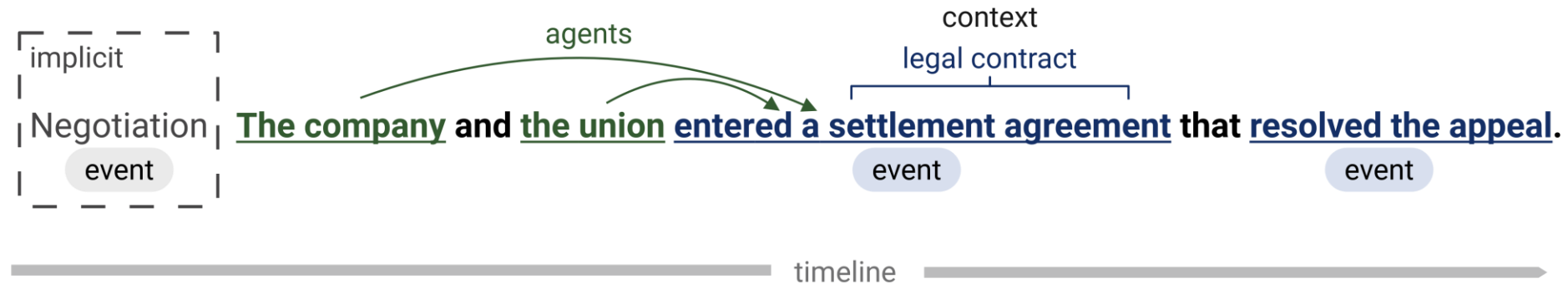
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In-context learning fails on complex reasoning tasks: arithmetic reasoning, symbolic reasoning, commonsense reasoning and temporal reasoning



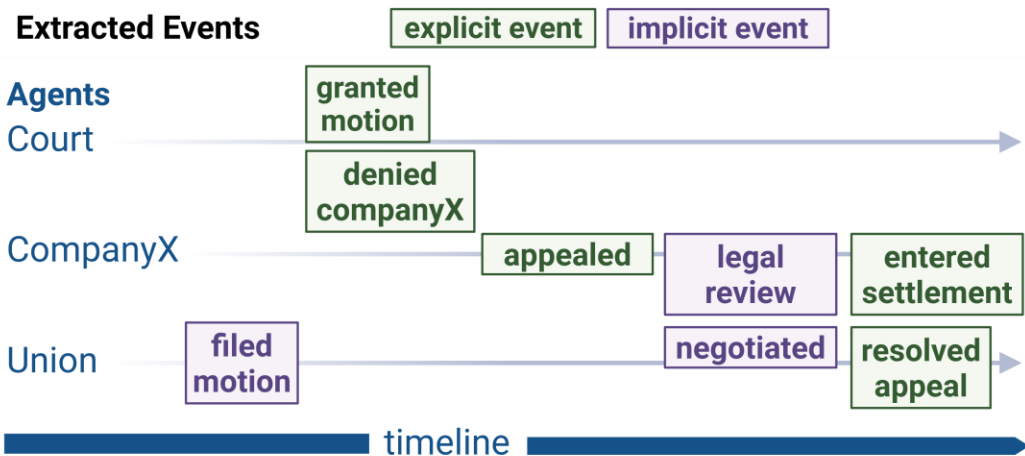
Event ordering: Understanding when legal events occur and how they relate.

Task overview: Event ordering on the LexTime dataset

Context paragraph

On November 25, 2020, the district court **granted** the unions summary judgment motion and **denied** CompanyX. CompanyX subsequently **appealed** the matter to the United States Court of Appeals for the Sixth Circuit (no. 20-xxxx). on September 17, 2021, however, the company and the union **entered** a settlement agreement that **resolved** the appeal.

Extracted Events



LexTime instance

Does the event
companyX **appeal**?

entering a settlement agreement **follow**

temporal relationship

Label

Yes

Event-centric reasoning:

- states/actions situated in time
- temporal relations (before / after / simultaneous)

Labor-related federal complaints (2020-2024)

512 manually annotated instances, balanced (yes/no labels)

Two event types:

- Explicit events: stated directly (e.g., *granted motion*)
- Implicit events: inferred from context (e.g., *negotiated*)

Query format:

{event A, temporal relation, event B}
→ “Does A happen before B?”

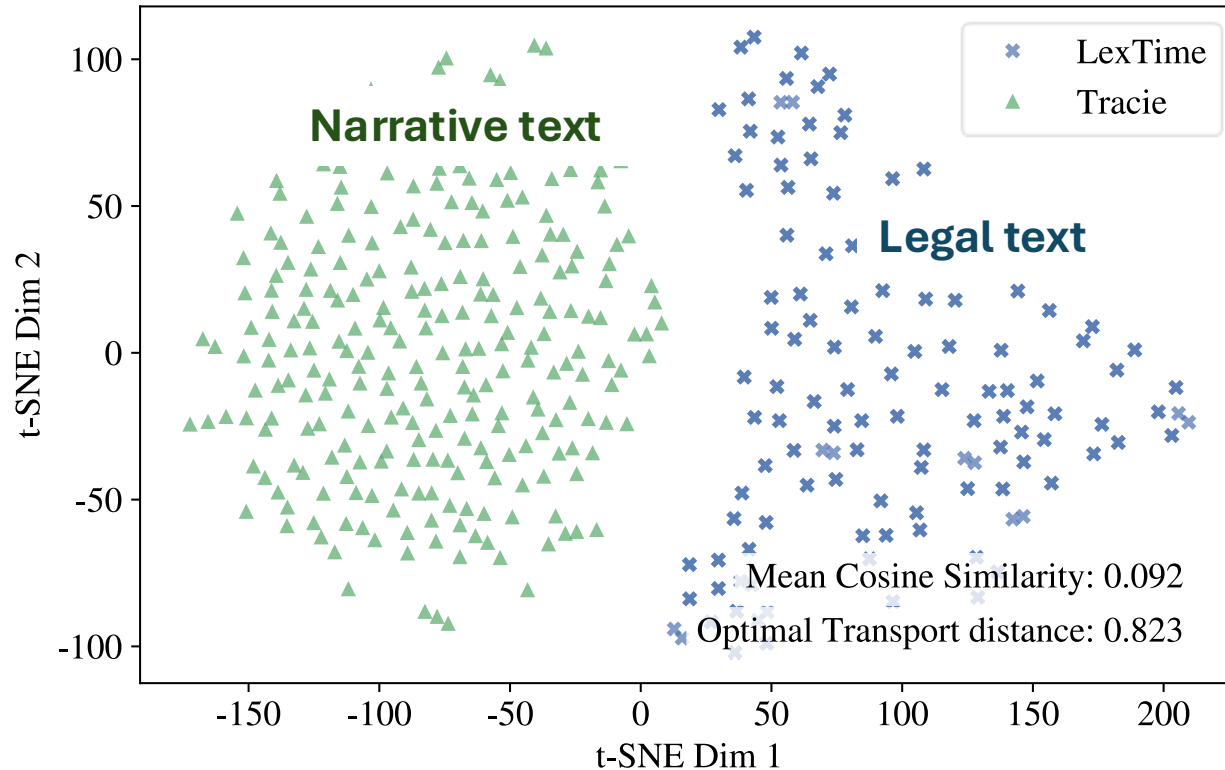
Comparative Dataset: TRACIE, narrative benchmark

Temporal Reasoning on Implicit Events from Distant Supervision

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Specialized language: Legal language is a linguistic genre and register



Visualization of the split of paragraph embeddings containing events using t-SNE (Van der Maaten and Hinton, 2008).

Low mean cosine similarity indicates that paragraphs are semantically different on average.

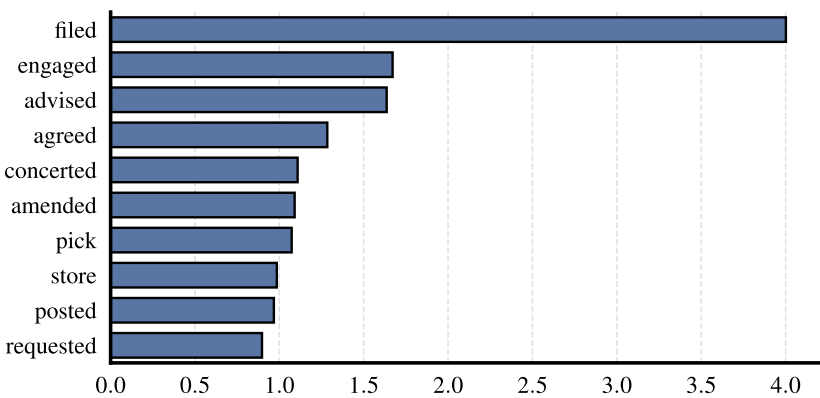
High optimal transport distance shows that the embedding distributions are distinct.

(RQ) How do language models react to the unique features of legal language for event ordering?

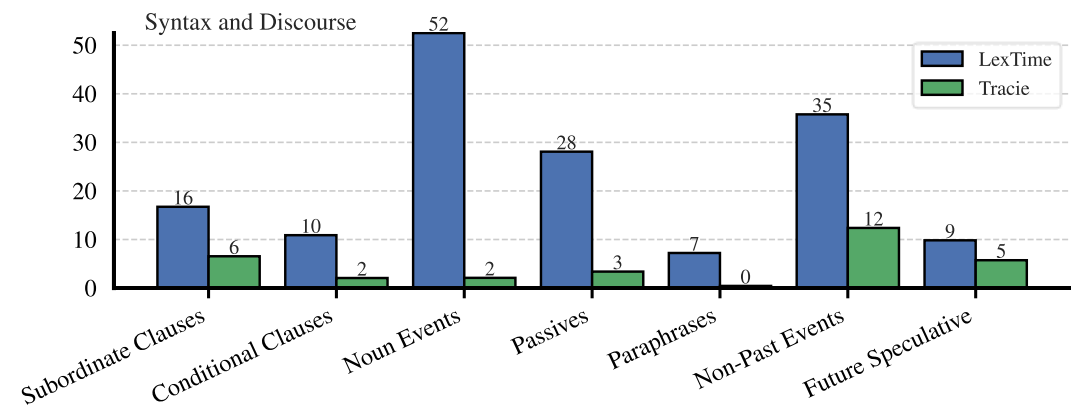
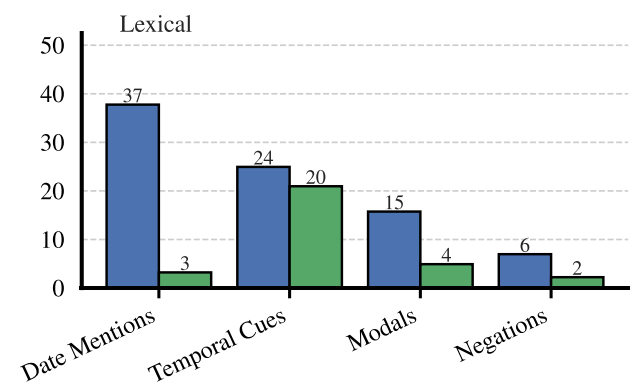
Legal language: Distinct distribution

Lexical and syntactic differences (LexTime vs. Tracie)

- Longer sentences (28 vs 9 tokens, median)
- Specialized legal vocabulary (*filed*, *granted*, *amended*)
- Higher event density (4.01 vs 1.60 events/sentence)
- More subordination (+10.5%) and modality (+11%)
- Frequent nominalization (+50.4%), paraphrasing (+6.56%), passive voice (+24.76%)

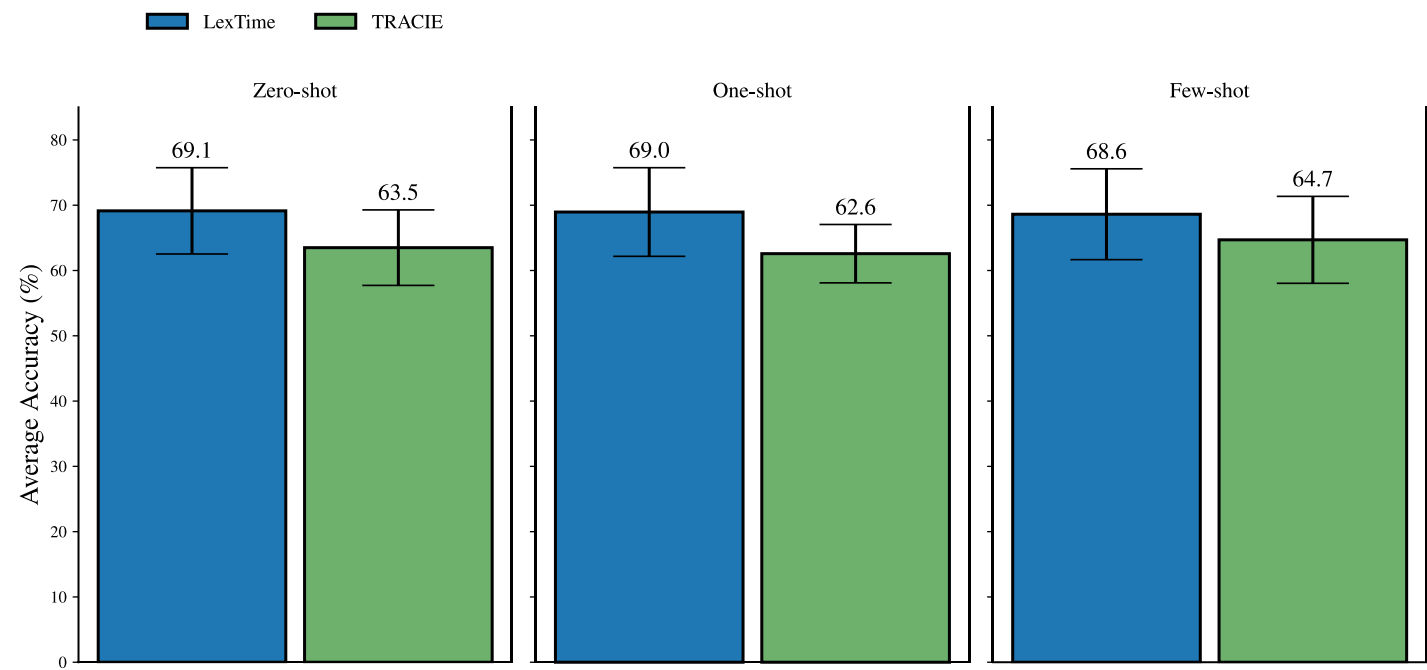


Top-10 event triggers in LexTime by frequency (%).



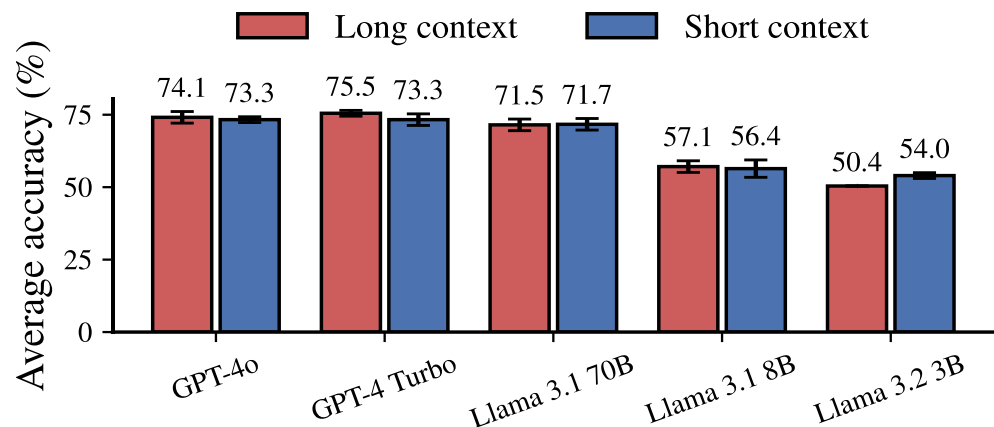
Comparison of linguistic features in LexTime and Tracie. Values represent the percentage of sentences that contain each features, indicating frequency in the text.

Result 1: Models achieve higher event ordering accuracy on LexTime than on TRACIE, by up to 10.5% (GPT-4o, few-shot)



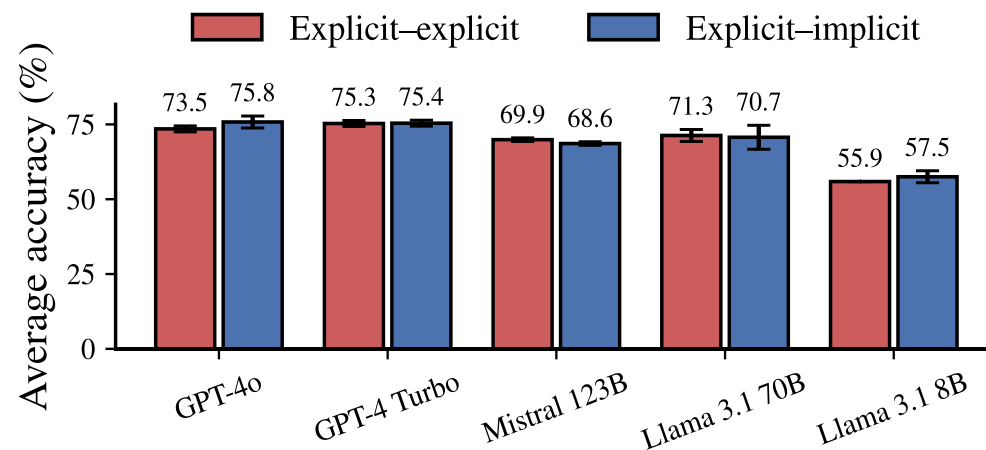
Results are averaged from GPT-4, GPT4-Turbo, Llama 3.1_{70B}, Llama 3.1_{8B} and Mistral_{123B}. 3 runs.

Result 2: Large models generally benefit from longer context windows



Context length (short <150 tokens, long ≥150).

Result 3: Large models perform better when implicit events are included



Explicit vs. implicit event pairs.

Query: Does event A happen before event B?

Example explicit-explicit pair: {explicit event A, TR, explicit event B}

Example explicit-implicit pair: {explicit event A, TR, implicit event B}

Results are averaged from zero-shot, one-shot, few-shot settings. 3 runs.

Result 4: Chain-of-thought prompting is not effective

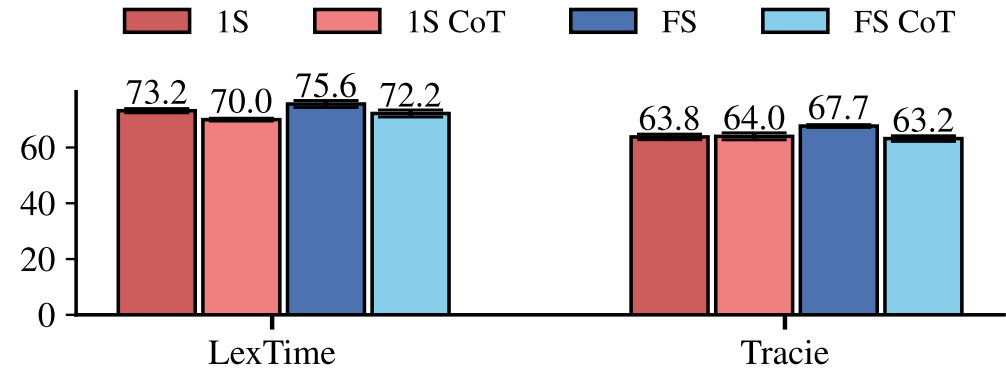
Example used for CoT prompts

{Context} *Before assigning the plaintiff to work, the defendant hired employees, including the plaintiff. The plaintiff was hired as a trackman.*

{Query} *Does hiring employees happens before assigning the plaintiff to work?*

{Answer} *Reasoning: 1. The defendant hired employees, including the plaintiff.
2. The plaintiff was hired as a trackman.
3. The employees including the plaintiff were all assigned to work.*

The correct answer is: "yes"



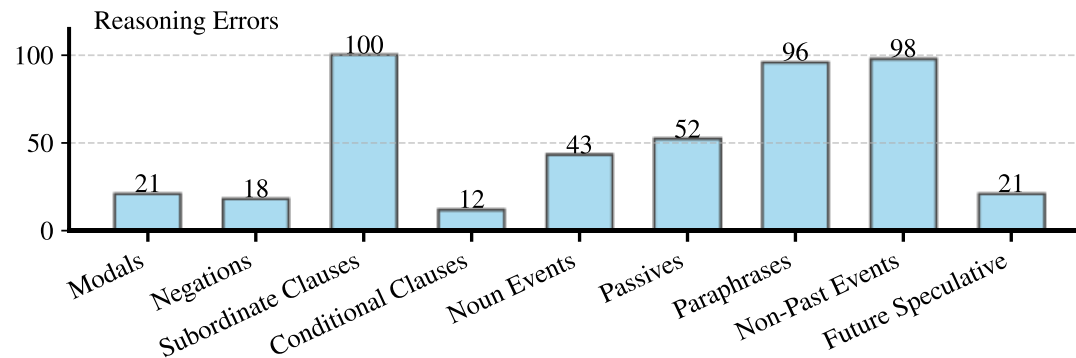
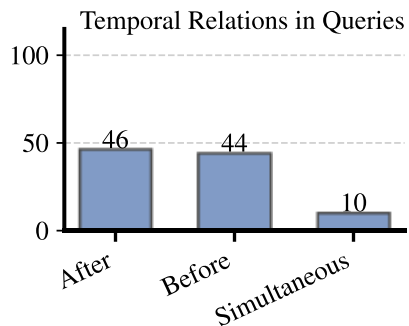
Performance gap (accuracy) with and without CoT prompting.

Results are averaged from GPT-4, GPT4-Turbo, Llama 3.1_{70B} and Mistral_{123B}. 3 runs.

Reasoning errors are associated with the linguistic specificities of legal language.

Compared to the overall dataset, errors are associated with:

- +89% increase in paraphrases
- +84% in subordinate clauses
- +63% in events not in the past simple tense
- +24% in passive voice
- +12% in negations and future speculative constructions
- +6% in modality markers
- +2% in conditional clauses.



Error analysis for LexTime on 100 misclassified samples from the few-shot setting across GPT-4, GPT-4 Turbo and Llama 3.1_{70B}.

Takeaways

Discussion & Implications

- LexTime introduces temporal reasoning to legal NLP.
- LLMs show improved performance on legal event ordering in LexTime compared to narrative, but linguistic complexities commonly found in legal language contribute to model errors.
- Calls for domain-sensitive modeling, tailored to specialized language types, and possibly data simplification.
- Opens directions for domain adaptation and cross-domain generalization.

Limitations & Future Work

- Only 512 samples and limited temporal relations schema (before/after/simultaneous).
- Focus on verb event triggers (noun event triggers are not annotated).
- Future work: extend to richer temporal structures, and across specialized domains.